

HOMEWORK 2

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 09/15/2023

Problem 1. Let $\{x_n\}$ be a sequence that converges to x^* with order $\alpha > 1$. Show that this implies that $\{x_n\}$ converges to x^* superlinearly; i.e., show that

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|} = 0.$$

Problem 2. Identify the limit and the order of convergence for the following sequences (if linear, provide the rate constant r as well).

1. $a_n = 2^{-2^n + n}$
2. $a_n = \sin\left(\frac{1}{2n}\right) - \frac{1}{2n}$.
3. $a_n = 2^{-n} + 100(3^{-n})$

Problem 3. Implement the bisection method (submit your code with this problem). Then, use it to obtain the solution to the following problems to an absolute error of 10^{-5} and starting on the provided interval:

1. $f(x) = \sin(x)$ starting on interval $[2, 4]$
2. $x^3 = 3^x$ starting on the interval $[-4, 4]$

State the required number of iterations in each case.

Problem 4. Implement Newton's method with the stopping criteria $|x_k - x_{k-1}| < 10^{-8}$ (submit your code with this problem). Then, use it on the function $f(x) = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ (which has only one root, $x = 0$) starting with an initial guess of $x_0 = 0.9$. Repeat this starting with an initial guess of $x_0 = 1.1$. What happens? Why?

Problem 5. Consider the function $f(x) = \text{sign}(x-a)\sqrt{|x-a|}$. Try to find the zero of the function at $x = a$ using Newton's method. What happens? Does this contradict the convergence theorem for Newton's method?

Hint: To calculate the derivative, compute it separately for $x < a$ (where $\text{sign}(x-a) = -1$ and $|x-a| = a-x$) and $x > a$ (where $\text{sign}(x-a) = 1$ and $|x-a| = x-a$).